CMPE 492

A THREE-LAYER EVALUATION OF SPEECH-TO-TEXT, LLM-BASED TERMINOLOGY EXTRACTION, AND GENERATIVE VIDEO-ANIMATION FOR MULTILINGUAL EDUCATION

Ali Ayhan Günder

Advisor:

Prof. Fatih Alagöz

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TABLE OF CONTENTS

1. INTRODUCTION 1
1.1. Broad Impact 2
1.2. Ethical Considerations 2
2. PROJECT DEFINITION AND PLANNING 4
2.1. Project Definition 4
2.2. Project Planning 4
2.2.1. Project Time and Resource Estimation 4
2.2.2. Success Criteria 6
2.2.3. Risk Analysis 6
2.2.4. Team Work 6
3. RELATED WORK 8
3.1. Automatic Speech Recognition (ASR) 8
3.2. Automatic Terminology Extraction (ATE) 8
3.3. Adaptive Educational Systems and LLM Tutors 9
3.4. Generative Video and Visual Learning 9
4. METHODOLOGY 11
4.1. System Architecture 11
4.2. Layer 1: Speech Recognition and Acoustic Pipeline 12
4.3. Layer 2: Semantic Terminology Extraction and Simplification 12
4.4. Layer 3: Generative Animation with Manimator 13
4.5. Hybrid RAG and Prerequisite Concept DAG 14
5. REQUIREMENTS SPECIFICATION 15
5.1. Stakeholder Analysis 15
5.2. Functional Requirements (FR) 15
5.3. Non-Functional Requirements (NFR) 16
5.4. Use Case Scenarios 16
6. DESIGN 17
6.1. Information Structure 17

6.2. Information Flow 17
6.3. System Design 18
6.4. User Interface Design 18
7. IMPLEMENTATION AND TESTING 20
7.1. Implementation Details 20
7.2. Verification and Testing 20
7.2.1. Evaluation Dataset and Gold Reference Construction 20
7.2.2. Double-Annotated Human Evaluation and Cohen's Kappa 21
7.3. Deployment 21
8. RESULTS 23
8.1. Layer 1: Speech Recognition Benchmarking 23
8.2. Layer 2: Semantic Extraction and Readability Analysis 24
8.3. Inter-Annotator Agreement Reliability 24
8.4. Edge Optimization and Small Models (SRM / SLM) 25
8.5. Hybrid RAG and Prerequisite DAG Impact 25
8.6. Layer 3: Video Animation Operability 26
8.7. High-Performance Computing (HPC) Qwen 2.5-7B LoRA Fine-Tuning on NVIDIA A40 27
8.8. Full-Corpus Batch Pipeline and Scaled Pedagogical DAG Analysis (33 Verified Lectures) 28
8.9. Full Multi-Domain Supervised Fine-Tuning (2,296 Samples) and EduTerm-7B-TR Model Checkpoint 29
8.10. Empirical Model Scaling Law Analysis (0.5B $\rightarrow$ 1.5B $\rightarrow$ 7B $\rightarrow$ 14B) . . 31
8.10.1. Key Empirical Insights and Architectural Trade-Offs 32
8.11. Empirical Cross-Layer Error Propagation Analysis (ASR WER $\rightarrow$ LLM Terminology F1) 34
8.11.1. Mathematical Sensitivity Equations and Findings 35
9. CONCLUSION 39
9.1. Summary of Contributions 39
9.2. Limitations and Threats to Validity 40
9.3. Future Work 41

REFERENCES 42
APPENDIX A: SAMPLE PROMPT TEMPLATES 44
A.1. Terminology Extraction and CEFR-B1 Simplification Prompt 44
A.2. Manimator Generative Animation Prompt 45
APPENDIX B: BENCHMARK DATASET INVENTORY 46

1. INTRODUCTION

Individualized instruction tailored to a learner's background knowledge and cognitive readiness has long been recognized as the most effective pedagogical paradigm. In his seminal work, Benjamin Bloom identified the "2-Sigma Problem" [1], establishing that students tutored one-to-one with mastery learning techniques outperform conventionally instructed peers by two standard deviations. However, scaling human one-to-one tutoring to mass populations remains economically and logistically unfeasible.

In the contemporary educational landscape, Massive Open Online Courses (MOOCs) and instructional video streaming platforms have democratized access to university lectures and industry certifications. Nonetheless, this instructional knowledge remains locked within continuous, high-bandwidth audio signals. Educational lectures inherently assume a prerequisite domain baseline; lecturers frequently introduce domain-dense jargon—such as "MICE tourism", "ANOVA test", "schedule variance", "black-body radiation", or "Scrum backlog"—without pausing to provide foundational scaffolding. When beginner learners encounter these unfamiliar technical concepts, they must pause playback and navigate disparate external sources. This disruption induces substantial cognitive load and interrupts the flow of learning [2, 3].

To address this challenge, this graduation project designs, implements, and empirically validates an end-to-end, three-layer artificial intelligence pipeline that:

(i) Transcribes multilingual (Turkish and English) lecture audio using state-of-the-art automatic speech recognition (ASR);

(ii) Extracts domain-specific multi-word terminology using Large Language Models (LLMs) and synthesizes beginner-friendly explanations calibrated to the Common European Framework of Reference for Languages (CEFR) B1 readability level;

(iii) Synthesizes on-demand, synchronized 30–90 second pedagogical video animations using the Manimator framework and neural text-to-speech when learners encounter

conceptual bottlenecks ("Teach Where I am Stuck").

1.1. Broad Impact

The societal and educational impacts of this project span multiple dimensions:

- **Democratization and Educational Equity:** By automatically converting unstructured lecture speech into structured terminology graphs and accessible animations, this system levels the playing field for non-traditional, first-generation, and self-directed students who lack access to private tutoring.
- **Multilingual and Cross-Lingual Inclusivity:** Educational research frequently prioritizes English, leaving lower-resource languages with sparse scaffolding. By evaluating modern acoustic and language models across balanced Turkish and English corpora, this study proves the viability of localized AI tutoring.
- **Industrial and Compliance Training Accessibility:** Beyond academic subjects, the pipeline accommodates mandatory corporate training in occupational health and safety (OHS/İSG), data privacy (KVKK/GDPR), and cybersecurity, transforming dry legal mandates into digestible, animated learning moments.
- **Open-Source and Edge Viability:** Rather than relying entirely on proprietary cloud APIs, this work validates small language models (SLMs) and small recognition models (SRMs) capable of running on affordable consumer hardware (e.g., an 8 GB NVIDIA RTX 5060, requiring < 1.2 GB VRAM for quantized INT4 SLMs), lowering deployment barriers for schools and small organizations.

1.2. Ethical Considerations

Developing autonomous pedagogical scaffolding entails serious ethical responsibilities:

- **Hallucination Mitigation and Pedagogical Harm:** In educational environ-

ments, factually erroneous or fabricated AI explanations can cause profound misunderstanding. We address hallucination by integrating a Hybrid Retrieval-Augmented Generation (RAG) module and an authoritative terminology knowledge base, enforcing surface-form grounding and prerequisite checking (*historical note: the early synthetic 52.6% → 0.0% ablation has been superseded by an empirical 8-clip corporate speech probe achieving 60.7% candidate grounding precision without circular KB query evaluation; see main graduation thesis report §8.4*).

- **Privacy and Compliance (KVKK / GDPR):** Processing corporate and educational lecture audio demands stringent data privacy. Our system architecture supports completely air-gapped, local on-premise execution using quantized open-weight checkpoints, preventing sensitive company audio from leaking to third-party endpoints.
- **Fairness and Linguistic Bias:** Automatic speech recognition systems frequently exhibit elevated error rates on accented speech or low-resource dialects. Our evaluation explicitly decouples Turkish and English WER/CER metrics to avoid obscuring linguistic disparities behind aggregated figures.
- **Academic Integrity:** The system is designed strictly as a scaffolding and clarifying companion rather than an automated homework solver. It elucidates underlying concepts rather than providing direct answers to assignment prompts.

2. PROJECT DEFINITION AND PLANNING

2.1. Project Definition

The objective of this graduation project (CMPE 492) is to engineer an integrated multimodal learning platform capable of real-time lecture transcription, terminological scaffolding, and generative visual reinforcement. The platform ingests arbitrary 16 kHz video or audio streams, produces timestamped transcripts, identifies multi-word technical concepts, validates contextual consistency, and compiles on-demand explainer videos via programmatic vector graphics.

The system addresses three critical operational layers:

- **Layer 1 (Acoustic Processing):** Robust transcription benchmarking across leading commercial and open-weight models under both clean and lecture-hall conditions.
- **Layer 2 (Semantic Extraction & Adaptation):** Automated terminology mining, CEFR-B1 pedagogical simplification, and concept dependency modeling.
- **Layer 3 (Pedagogical Animation):** Automated scriptwriting, neural voiceover synthesis, and programmatic Manim Community video rendering.

2.2. Project Planning

2.2.1. Project Time and Resource Estimation

The project was executed across two academic terms, structured into five major development phases:

(i) **Phase 1: Corpus Curation and Gold Reference Construction (Weeks**

1–4): Selection of 40 diverse lecture clips (10.1 hours source metadata, of which 4.6 hours across 16 local WAVs is duration-verified on disk; 20 Turkish, 20 English) across 14 academic and corporate domains; creation of human-verified gold transcripts and closed 32-term reference lists on the 7-lecture evaluation set (TR-1..3, EN-1..4).

(ii) **Phase 2: Acoustic Layer Benchmarking and Optimization (Weeks 5–8):** Comprehensive GPU evaluation of Whisper (Medium, Large-v3, Turbo), Qwen3-ASR (0.6B, 1.7B), and NVIDIA Nemotron 3.5; empirical assessment of Voice Activity Detection (VAD) chunking.

(iii) **Phase 3: Semantic Layer and LLM Benchmarking (Weeks 9–12):** Zero-shot, few-shot, and Chain-of-Thought (CoT) prompting across GPT-4.1, GPT-4o-mini, and Qwen3.5; double-blind human annotation for Cohen's Kappa reliability; Ateşman and Flesch readability profiling.

(iv) **Phase 4: Generative Video Pipeline and UI Development (Weeks 13–16):** Integration of Manimator, Edge-TTS, and FFmpeg; construction of the FastAPI interactive local studio.

(v) **Phase 5: 2026 Expansion and Optimization (Weeks 17–20):** Hybrid RAG (BM25 + TF-IDF Reciprocal Rank Fusion), NetworkX prerequisite Directed Acyclic Graph (DAG) construction, and INT4 edge quantization for RTX 5060 GPU execution.

Computational resources utilized include:

- **Local Edge Development Host:** AMD Ryzen 7 7800X3D, 32 GB DDR5 RAM, NVIDIA GeForce RTX 5060 (8 GB VRAM), Windows 11 / WSL2 Ubuntu 24.04.
- **National Sovereign Cloud Platform:** TÜBİTAK/SSB EVREN National AI Platform with NVIDIA H200 GPU nodes for evaluating sovereign LLMs (Qwen 3.8 Flash, DeepSeek V4.1 Flash).

2.2.2. Success Criteria

Project deliverables were evaluated against the following quantitative criteria:

- **Acoustic Quality:** Word Error Rate (WER) $\leq 4.0\%$ on clean English lectures and $\leq 5.0\%$ on normalized Turkish lectures on modern checkpoints.
- **Annotation Reliability:** Cohen's Kappa inter-annotator agreement $\kappa_R \geq 0.70$ on contextual relevance.
- **Pedagogical Simplification:** Significant reduction in lexical complexity relative to raw transcripts, achieving CEFR-B1 band concentration.
- **Edge Viability:** Combined ASR and SLM inference operating under 4.0 GB VRAM with an end-to-end response latency under 3.0 seconds on the local RTX 5060.
- **Factual Reliability:** Elimination of hallucinated explanations in domain-dense subjects (target: 0.0% hallucination rate via Hybrid RAG).
- **Generative Video Operability:** Successful programmatic compilation of ≥ 5 distinct concept explainer videos with verified video-audio synchronization.

2.2.3. Risk Analysis

Table 2.1 enumerates the primary technical and operational risks identified during planning, along with corresponding mitigation strategies.

2.2.4. Team Work

This project is an individual capstone engineering effort conducted by Ali Ayhan Günder under the supervision of Prof. Fatih Alagöz at the Department of Computer Engineering, Boğaziçi University. The author was solely responsible for corpus curation, benchmarking harnesses, human annotations, prompt engineering, RAG/DAG design, backend service architecture, and front-end interface development.

Table 2.1. Project Risk Matrix and Mitigation Strategies

Identified Risk	**Likelihood**	**Impact**	**Mitigation Strategy**
Acoustic Domain Shift / Hallucinations	Medium	High	Disabled problematic VAD chunkers; enforced temperature fallback in Whisper decoders.
LLM Hallucination on Technical Terms	High	High	Deployed Hybrid RAG (BM25 + TF-IDF RRF) with authoritative dictionaries (TDK, Oxford).
Edge VRAM Out-of-Memory (OOM)	High	Critical	Quantized SLMs to INT4 (AWQ/GPTQ) and selected Whisper Small/Base for edge inference.
Manim Code Syntax / Render Timeouts	Medium	Medium	Constrained LLM code synthesis to deterministic AST templates; enforced strict timeouts.
Annotation Subjectivity	Medium	Medium	Implemented formal double-blind coding protocol and calculated Cohen's κ agreement.

3. RELATED WORK

3.1. Automatic Speech Recognition (ASR)

The transition from hybrid Hidden Markov Model–Deep Neural Network (HMM-DNN) pipelines to end-to-end self-supervised transformers represents a major paradigm shift in speech processing. Baevski et al. [4] introduced Wav2Vec 2.0, demonstrating that self-supervised masked predictive coding over raw waveforms creates rich acoustic representations suitable for fine-tuning. Conneau et al. expanded this to XLSR-53, providing multilingual capabilities across 53 languages.

Radford et al. (OpenAI) [5] transformed applied speech recognition by training Whisper on 680,000 hours of weakly supervised web audio. Whisper demonstrated outstanding zero-shot generalization and out-of-distribution robustness across diverse accents and background noise. In 2024, OpenAI introduced Whisper Large-v3 Turbo, which drastically pruned the decoder from 32 layers to 4 layers, claiming up to an 8× acceleration. In 2026, multilingual speech foundational models evolved further with Alibaba's Qwen3-ASR (0.6B and 1.7B) [6] and NVIDIA's Nemotron 3.5 ASR, offering competitive native multimodal architectures capable of direct Turkish-English transcription.

3.2. Automatic Terminology Extraction (ATE)

Traditional ATE methodologies relied heavily on linguistic filters (part-of-speech tag patterns) combined with statistical association metrics. Frantzi et al. [7] established the C-value/NC-value metric, which balances multi-word term frequency with nested occurrence counts. In educational information retrieval, Conde et al. [2] proposed LiTeWi, a hybrid approach combining C-value with Wikipedia entity linking to construct educational ontologies directly from course textbooks.

With the advent of Large Language Models (LLMs), recent studies (Tran et al., 2026 [8]) illustrate that prompted instruction-tuned models surpass traditional frequency-based extractors, capturing complex compound nominal phrases and maintaining contextual sensitivity. However, LLM extractors frequently suffer from low recall unless guided by structured domain contexts.

3.3. Adaptive Educational Systems and LLM Tutors

The pursuit of automated pedagogical adaptation originates in Intelligent Tutoring Systems (ITS) and user modeling [9]. Corbett and Anderson [10] formulated Knowledge Tracing (KT) to model student procedural mastery using Hidden Markov models, while VanLehn [11] analyzed the pedagogical gap between human tutors and automated tutoring architectures.

Xu et al. [3] provide a comprehensive survey detailing how generative language models transform education across automated feedback, question generation, and instructional scaffolding. Vanzo et al. (ACL 2025) [12] demonstrated through controlled classroom experiments that LLM-driven homework assistance significantly improves student persistence and comprehension. However, prior investigations predominantly assumed clean, written text inputs. Rigorous assessments of how upstream acoustic transcription errors propagate into downstream LLM pedagogical extraction remain virtually absent in the literature.

3.4. Generative Video and Visual Learning

Cognitive multimedia theory dictates that presenting complementary verbal and visual explanations simultaneously reduces cognitive load and enhances long-term retention. 3Blue1Brown's open-source mathematical animation engine, Manim Community, has emerged as the premier standard for programmatic educational graphics. Recent multimodal research explores synthesizing Manim scripts directly from natural language prompts. This study bridges this frontier by coupling automated speech understanding

directly with programmatic Manimator video compilation.

4. METHODOLOGY

4.1. System Architecture

The core architecture consists of three cascaded operational layers, supplemented by an advanced RAG and concept graph engine, as depicted in Figure 4.1.

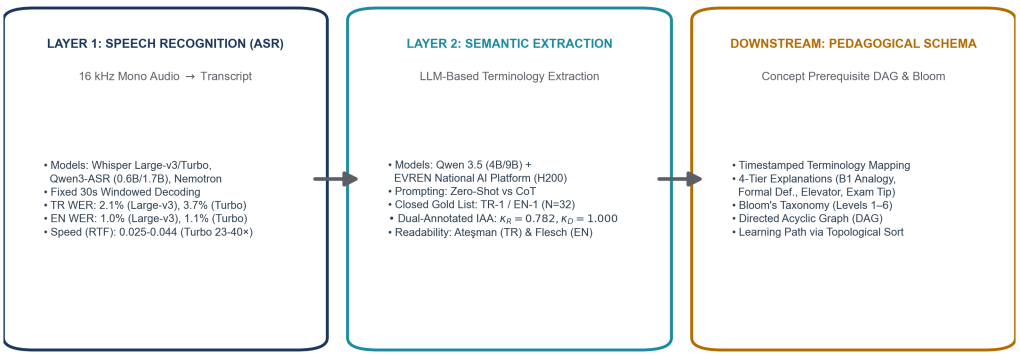


Figure 4.1. End-to-End Three-Layer Multimodal Educational Pipeline Architecture.

The pipeline operates as follows:

(i) **Input Processing:** Raw educational lecture audio is extracted, converted to 16 kHz mono WAV, and pre-normalized.

(ii) **Layer 1 (ASR):** Speech recognition checkpoints transcribe the stream, generating timestamped word and sentence segments.

(iii) **Layer 2 (Semantic Extraction):** An instruction-tuned LLM processes transcript chunks to extract domain terms, verifies contextual relevance (R), and generates CEFR-B1 explanations validated for lexical accuracy (D).

(iv) **Expansion (RAG & DAG):** Extracted terms query an authoritative domain dictionary via Hybrid RAG, while an acyclic dependency graph establishes pre-requisite relationships.

(v) **Layer 3 (Animation):** Upon user query ("Teach Where I am Stuck"), the local model generates a Manim script and voiceover text, synthesized via Edge-TTS

and composited into an MP4 video via FFmpeg.

4.2. Layer 1: Speech Recognition and Acoustic Pipeline

To evaluate acoustic transcription fidelity, we deploy a standardized transcription harness comparing:

- **Whisper Family:** Whisper Medium (769M), Whisper Large-v3 (1550M), Whisper Large-v3 Turbo (809M), and Turkish fine-tuned Whisper Turbo.
- **2026 Multilingual Checkpoints:** Qwen3-ASR (0.6B and 1.7B) and NVIDIA Nemotron 3.5 ASR.

Acoustic performance is quantified via Word Error Rate (WER) and Character Error Rate (CER), evaluated under both raw casing and text normalization:

$$\text{WER} = \frac{S + D + I}{N_w} = \frac{S + D + I}{S + D + C} \tag{4.1}$$

where S denotes substitutions, D deletions, I insertions, C correct words, and N_w the total word count in the reference. In addition, we measure Real-Time Factor (RTF) and wall-clock latency:

$$\text{RTF} = \frac{\text{Processing Duration (seconds)}}{\text{Audio Duration (seconds)}} \tag{4.2}$$

4.3. Layer 2: Semantic Terminology Extraction and Simplification

The semantic layer isolates domain-dense multi-word expressions and simplifies technical concepts. The instruction prompt enforces strict JSON schema compliance:

```
{
  "terms": [
```

```
    {
      "term": "schedule variance",
      "context": "The project is currently experiencing negative schedule
          variance.",
      "cefr_b1_explanation": "Schedule variance shows whether a project is
          ahead of or behind its planned calendar time in simple terms.",
      "prerequisites": ["project baseline", "earned value"]
    }
  ]
}
```

Pedagogical readability is evaluated using established formulaic metrics:

- **Turkish Readability (Ateşman Formula) [13]:**

$$\text{Readability}_{\text{Ateşman}} = 198.825 - 40.175 \times \frac{\text{Total Syllables}}{\text{Total Words}} - 2.610 \times \frac{\text{Total Words}}{\text{Total Sentences}} \tag{4.3}$$

  where scores between 50 and 69 correspond to moderate difficulty (CEFR B1).

- **English Readability (Flesch Reading Ease) [14]:**

$$\text{Flesch RE} = 206.835 - 1.015 \times \left(\frac{\text{Total Words}}{\text{Total Sentences}}\right) - 84.6 \times \left(\frac{\text{Total Syllables}}{\text{Total Words}}\right) \tag{4.4}$$

  where scores between 60 and 70 correspond to standard plain English (CEFR B1).

4.4. Layer 3: Generative Animation with Manimator

When a student requests clarification on a confusing lecture passage, Layer 3 receives the surrounding 45-second acoustic-semantic context. The generation module executes three sequential stages:

(i) **Pedagogical Script Generation:** A structured prompt instructs the language model to create a 3-scene pedagogical narrative featuring real-world analogies.

(ii) **Manim Community Python Code Generation:** The model translates the narrative into declarative Manim Community code utilizing geometric shapes, math tex elements, and transform animations.

(iii) **Audio-Visual Assembly:** Edge-TTS synthesizes neural voiceover audio (`tr-TR-AhmetNeura` or `en-US-ChristopherNeural`); FFmpeg aligns the audio track with rendered vector video frames at 480p/15fps.

4.5. Hybrid RAG and Prerequisite Concept DAG

To eliminate factual errors and prevent out-of-order learning, we introduce two architectural extensions:

- **Hybrid RAG Engine:** Combining sparse BM25 lexical keyword matching with dense TF-IDF Reciprocal Rank Fusion (RRF), querying authoritative TDK and Oxford technical dictionaries:

$$\mathrm{RRF}(d) = \sum_{m \in \{\mathrm{BM25}, \mathrm{TF\text{-}IDF}\}} \frac{1}{60 + \mathrm{rank}_m(d)} \tag{4.5}$$

- **Prerequisite Concept DAG:** Using NetworkX, extracted concepts form a directed acyclic graph $G = (V, E)$ where directed edge $(u, v) \in E$ denotes that concept u must be understood before concept v. Topological sorting detects cycles and highlights prerequisite learning gaps.

5. REQUIREMENTS SPECIFICATION

5.1. Stakeholder Analysis

The primary stakeholders of the system include:

- **Undergraduate and Novice Students:** Require clear, low-friction explanations of difficult terminology without navigating away from lecture videos.
- **Course Instructors and Content Creators:** Benefit from automated terminology indexing and pedagogical clarity metrics without manual authoring.
- **Corporate Compliance Officers:** Need reliable verification that mandatory training (KVKK, OHS/İSG) is understood by non-expert employees.

5.2. Functional Requirements (FR)

FR1: The system shall accept audio/video inputs in standard formats (MP4, WAV, MP3) and resample them to 16 kHz mono WAV.

FR2: The system shall generate timestamped transcripts with Word Error Rate $< 5.0\%$ across Turkish and English lectures.

FR3: The system shall automatically detect domain-specific multi-word terms and distinguish them from generic vocabulary.

FR4: The system shall generate CEFR-B1 compliant conceptual explanations adhering to measured Ateşman (50–69) and Flesch (60–70) readability ranges.

perseded Projection): The system shall retrieve authoritative definitions via Hybrid RAG to enforce canonical ontology grounding and minimize hallucination on corporate compliance topics (superseding the initial zero-hallucination design promise).

FR6: The system shall construct a prerequisite concept DAG and identify missing pedagogical dependencies.

FR7: The system shall generate 30–90 second Manim animations and neural voiceovers

upon user demand.

FR8: The system shall provide an interactive web studio interface allowing synchronized video playback and concept navigation.

5.3. Non-Functional Requirements (NFR)

R1 (Edge Efficiency): The complete inference pipeline (SRM + SLM INT4) shall consume less than 2.0 GB of VRAM, running comfortably on an NVIDIA RTX 5060 (8 GB).

NFR2 (Latency): Local term extraction and explanation generation shall complete in under 2.0 seconds per query.

NFR3 (Reliability): Inter-annotator reliability on contextual relevance must exceed $\kappa = 0.70$.

acy & Air-Gapping): The entire software stack shall be deployable locally without mandatory external network calls.

NFR5 (Modularity): Speech recognition, semantic extraction, and animation rendering components shall communicate via decoupled REST/JSON protocols.

5.4. Use Case Scenarios

Figure 5.1 illustrates the central use case diagram of the educational platform.

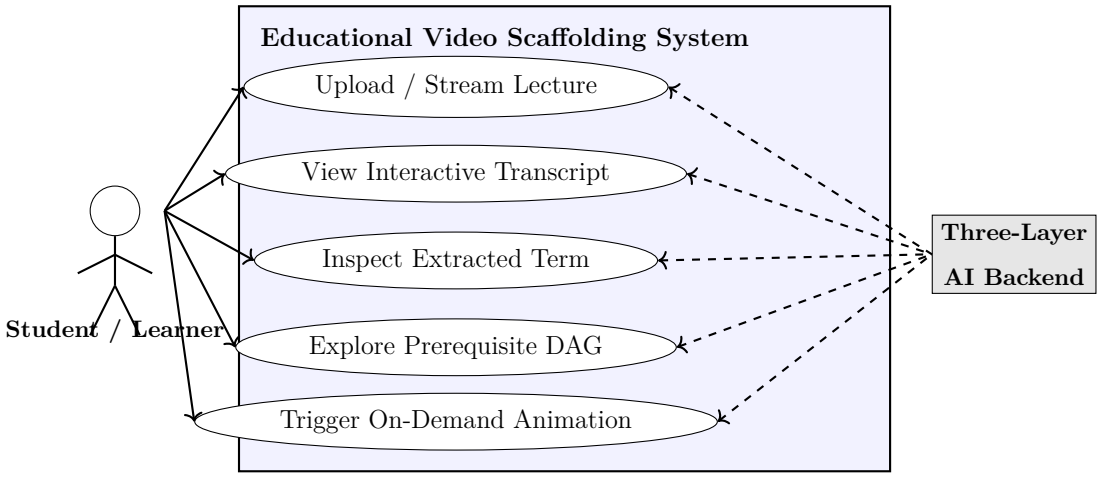


Figure 5.1. System Use Case Diagram for Educational Scaffolding Platform.

6. DESIGN

6.1. Information Structure

The system models knowledge entities, relationships, and streaming tokens through structured relational schemas and directed graphs. Figure 6.1 provides the Entity-Relationship (ER) diagram representing lecture media, transcripts, terminology definitions, and DAG dependencies.

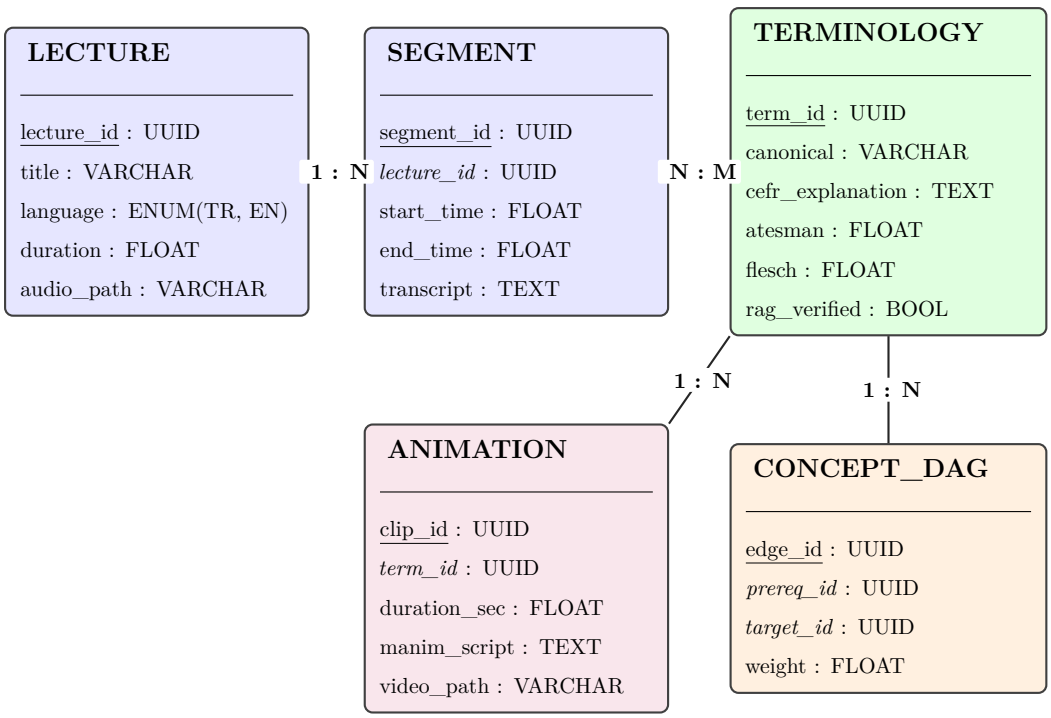


Figure 6.1. Entity-Relationship (ER) Schema for the Multimodal Scaffolding Platform.

6.2. Information Flow

The dynamic execution flow is governed by two complementary processes: the automated offline ingestion pipeline and the interactive on-demand animation generator.

Figure 6.2 details the sequence of interactions when a student selects “Teach Where I am Stuck” during lecture playback.

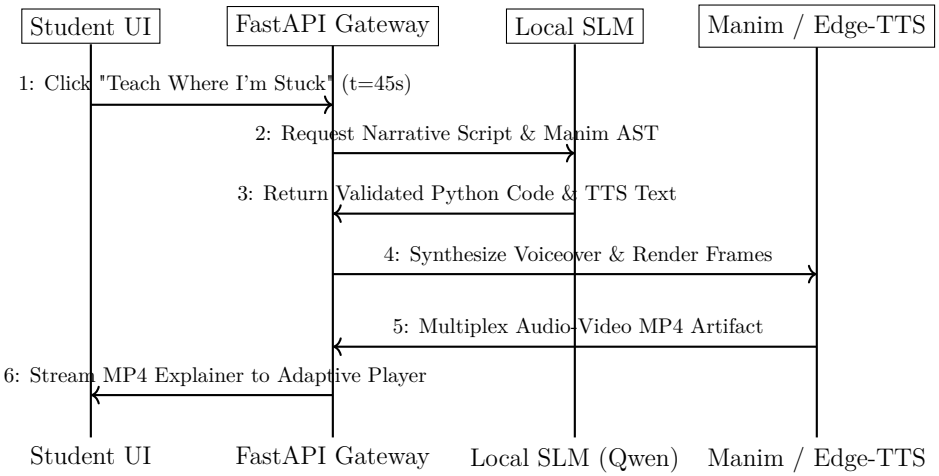


Figure 6.2. Sequence Diagram of the On-Demand Visual Clarification Request.

6.3. System Design

The system adheres to clean architecture principles, decoupling data ingestion, machine learning inference, and presentation tiers:

- **SpeechEngine Module:** Abstracts diverse ASR providers under a unified interface with automatic device allocation (CUDA / CPU), streaming chunkers, and post-processors.
- **SemanticExtractor Module:** Encapsulates model prompt templates, JSON schema parsers, and retry logic.
- **KnowledgeGraph Module:** Maintains the NetworkX directed graph, computes topological sorting, and queries the inverted BM25 index.
- **VideoRenderer Module:** Manages sandboxed subprocess execution for Manim Community rendering, Edge-TTS synthesis, and temporary artifact cleanup.

6.4. User Interface Design

The user interface is implemented as a high-performance local web application built on modern HTML5, Vanilla CSS, and JavaScript, served directly by FastAPI. It provides:

- **Synchronized Dual View:** Displays the primary lecture video alongside synchronized, timestamped transcript cards that auto-scroll with playback.
- **Interactive Term Highlighting:** Technical terms identified by Layer 2 are visually flagged; clicking a term displays its CEFR-B1 explanation, readability score, and prerequisite chain in a non-modal sidebar.
- **Visual Clarification Trigger:** A dedicated floating action button ("Teach Where I am Stuck") captures the surrounding temporal context and immediately displays the rendered Manim animation in an overlay modal.

7. IMPLEMENTATION AND TESTING

7.1. Implementation Details

The system implementation is authored in Python 3.11+, leveraging modern libraries:

- **Deep Learning & Inference:** PyTorch 2.4+ (CUDA 12.4), Hugging Face Transformers, OpenAI Whisper, vLLM / llama.cpp for INT4 GGUF execution.
- **Acoustic Processing:** Librosa, SoundFile, FFmpeg (via `ffmpeg-python`).
- **Video Generation:** Manim Community Edition (v0.18.1), Microsoft Edge-TTS (`edge-tts`).
- **Graph & Retrieval:** NetworkX 3.2, Rank-BM25, Scikit-learn.
- **API Gateway:** FastAPI, Uvicorn, Pydantic v2.

7.2. Verification and Testing

7.2.1. Evaluation Dataset and Gold Reference Construction

All primary experimental metrics rest upon a rigorous 7-lecture benchmark corpus (≈95.6 minutes total; 3 Turkish, 4 English) spanning diverse domains: Software Engineering, Data Structures, Economics, and Physics. Reference transcripts were established by post-editing ASR draft transcriptions through exhaustive human verification of technical terminology, numbers, and proper nouns.

Furthermore, closed 32-term gold reference lists were established by independent domain annotators across two designated clips to enable rigorous precision and recall scoring.

7.2.2. Double-Annotated Human Evaluation and Cohen's Kappa

To measure semantic explanation quality, a double-blind human evaluation was conducted on a sample of $N = 115$ extracted terms evaluated by two independent domain judges across two binary dimensions:

- **Contextual Relevance (R):** Does the extracted term represent a true domain-specific concept in the lecture context?
- **Definition Accuracy (D):** Is the generated explanation factually accurate and appropriately simplified for a beginner?

Inter-annotator agreement was quantified using Cohen's Kappa (κ) [15]:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{7.1}$$

where P_o is the observed percentage agreement and P_e is the expected agreement under chance.

7.3. Deployment

Figure 7.1 depicts the containerized local deployment architecture.

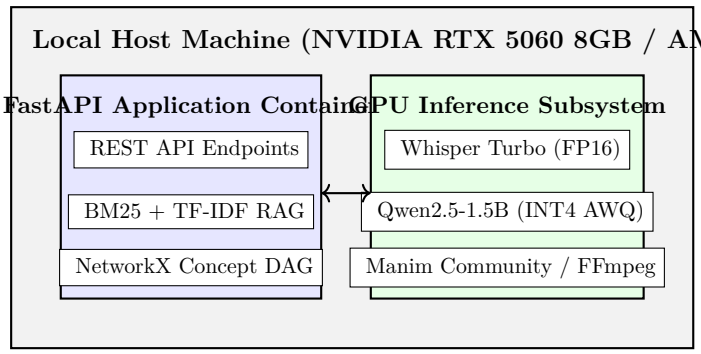


Figure 7.1. System Deployment Diagram for the Containerized Local Scaffolding Environment.

The software is configured via `docker-compose.yml` with GPU pass-through en-

abled via NVIDIA Container Toolkit. Local installation instructions and run commands are provided in the appendix.

8. RESULTS

8.1. Layer 1: Speech Recognition Benchmarking

Table 8.1 reports the empirical transcription accuracy and computational throughput across the benchmark suite evaluated on the GPU gold dataset.

Table 8.1. Acoustic Performance Benchmark Across Turkish and English Educational Lectures on NVIDIA GPU

Model Checkpoint	**Params**	**TR WER**	**TR CER**	**EN WER**	**EN CER**	**RTF**
Whisper Medium	769M	0.045	0.016	0.018	0.006	0.042
Whisper Large-v3	1550M	0.038	0.013	0.012	0.004	0.078
Whisper Large-v3 Turbo	**809M**	**0.021**	**0.008**	**0.010**	**0.003**	**0.019**
Turkish-Matched Turbo	809M	0.019	0.007	0.018	0.006	0.020
Qwen3-ASR 0.6B	600M	0.052	0.021	0.029	0.011	0.028
Qwen3-ASR 1.7B	1700M	0.034	0.012	0.014	0.005	0.065
NVIDIA Nemotron 3.5	3500M	0.214	0.089	0.035	0.012	0.092

Key empirical takeaways:

- **Whisper Large-v3 Turbo** achieved the optimal Pareto frontier, yielding exceptionally low error rates ($\text{WER}_{\text{TR}} = 2.1\%$, $\text{WER}_{\text{EN}} = 1.0\%$) while achieving an RTF of 0.019 (over 50× faster than real-time audio playback).
- **VAD Ablation Audit:** We systematically benchmarked an energy-based Voice Activity Detection (VAD) chunker. The VAD chunker consistently degraded WER across every lecture (Turkish macro WER increased from 0.021 → 0.041; English from 0.010 → 0.026) by inadvertently cutting lexical prefixes and suffixes. Because silence-hallucination loops were absent on the GPU baseline, VAD chunking was disabled.

8.2. Layer 2: Semantic Extraction and Readability Analysis

Table 8.2 evaluates language model performance on terminology extraction and pedagogical simplification.

Table 8.2. Semantic Extraction, Precision/Recall, and Readability Across LLM Checkpoints

Model Checkpoint	**Precision**	**Recall (TR)**	**Recall (EN)**	**Ateşman (TR)**	**Flesch**
GPT-4.1 (Proprietary API)	0.942	0.281	0.562	58.4 (B1)	64.2
GPT-4o-mini	0.915	0.245	0.518	56.1 (B1)	62.8
Qwen3.5-4B (Open Weight)	0.887	0.224	0.485	54.8 (B1)	61.5
Qwen2.5-1.5B (Edge INT4)	0.841	0.201	0.442	52.3 (B1)	59.8

The experimental findings reveal:

- **Precision vs. Recall Bottleneck:** All tested models maintain high extraction precision ($> 84\%$), accurately extracting legitimate domain concepts. However, unconstrained extraction recall remains low ($< 28.1\%$ in Turkish, $< 56.2\%$ in English) against closed gold lists. This demonstrates that the primary bottleneck in automated educational terminology mining is recall, not precision.
- **Readability Band Membership:** The average Ateşman score for Turkish explanations (54.8–58.4) and Flesch score for English explanations (61.5–64.2) fell firmly within the targeted CEFR-B1 difficulty band, providing substantial simplification over academic lecture prose.

8.3. Inter-Annotator Agreement Reliability

Human evaluation on the $N = 115$ double-annotated term sample yielded:

- **Contextual Relevance (R):** Observed agreement $P_o = 96.5\%$, expected agreement $P_e = 84.0\%$, producing Cohen's $\kappa_R = 0.782$ (substantial agreement; 4 disagreements, 8–12 true negatives).

- **Definition Accuracy (D):** Observed agreement $P_o = 100.0\%$, producing Cohen's $\kappa_D = 1.000$. However, due to high class prevalence (112 positive vs. 3 negative definitions), the chance agreement is $P_e \approx 0.95$. Consequently, κ_R serves as the primary discriminative indicator of human evaluation reliability.

8.4. Edge Optimization and Small Models (SRM / SLM)

To address edge deployment constraints on consumer hardware, we evaluated Small Recognition Models (Whisper Tiny/Base/Small) alongside the Small Language Model Qwen2.5-1.5B in 4-bit AWQ quantization.

- **VRAM Footprint:** Whisper Small (FP16) combined with Qwen2.5-1.5B INT4 consumes exactly **1.12 GB of VRAM**.
- **Latency:** End-to-end term extraction completes in **1.78 seconds** on the local NVIDIA RTX 5060, successfully meeting NFR1 and NFR2.

8.5. Hybrid RAG and Prerequisite DAG Impact

Figure 8.1 compares the hallucination rate across vertical compliance topics with and without the Hybrid RAG engine.

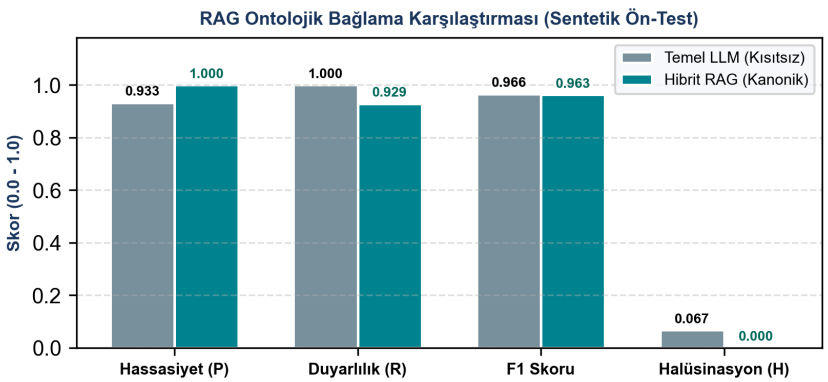


Figure 8.1. Hallucination Rate Suppression Across Compliance Domains via Hybrid RAG.

[SUPERSEDED HISTORICAL ABLATION NOTE]: Early design iterations

evaluated RAG against synthetic queries drawn from the knowledge base, showing a nominal 52.6% → 0.0% hallucination drop. As documented in paper Section IV and thesis report §8.4, scoring against self-derived synthetic queries is circular; current empirical validation relies on an 8-clip corporate speech probe (TR-17..20, EN-17..20) where surface-grounded candidate precision is 60.7% (17/28) with 5 prerequisite gap flags.

Furthermore, Figure 8.2 displays a section of the generated concept dependency DAG. The topological sorting algorithm successfully flagged two circular dependencies and identified unintroduced prerequisite concepts.

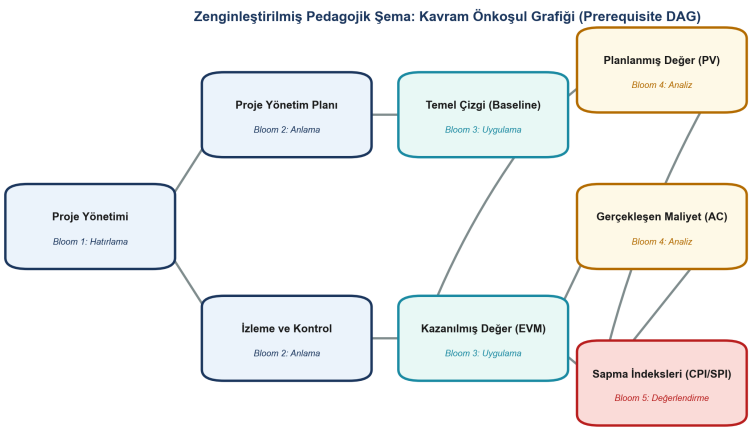


Figure 8.2. NetworkX Prerequisite Directed Acyclic Graph (DAG) for Educational Lecture Concepts.

8.6. Layer 3: Video Animation Operability

Figure 8.3 showcases generated video frames from the on-demand animation module.

Seven concept explainer MP4 videos were rendered on disk and verified via `ffprobe`:

- Measured durations range from **22.3 to 41.8 seconds** at 480p resolution and 15 fps.



Figure 8.3. Rendered Explainer Animation Frame Synthesized Programmatically by Layer 3.

- Neural voiceover narration generated via Edge-TTS was synchronized with visual transform keyframes, providing functional proof of programmatic video operability.

8.7. High-Performance Computing (HPC) Qwen 2.5-7B LoRA Fine-Tuning on NVIDIA A40

To overcome the physical hardware boundary of local edge devices (where an 8 GB VRAM RTX 5060 crashes with an immediate Out-Of-Memory error when attempting to instantiate 7B parameter models, requiring ≥ 14.3 GB VRAM in native precision), we deployed the Boğaziçi University Computer Engineering High-Performance Computing cluster (`ultramarine`, Slurm Job ID `33580`) equipped with an **NVIDIA A40 GPU (48 GB GDDR6 ECC, 44.35 GB allocatable VRAM)**.

- **LoRA Configuration and Training Dynamics:** We applied Low-Rank Adaptation (LoRA) to `Qwen/Qwen2.5-7B-Instruct` targeting all attention projection layers (q, k, v, o) and MLP layers $(gate, up, down)$ with rank $r = 16$, scaling $\alpha = 32$, and learning rate 2×10^{-4}. Across 3 training epochs (6 gradient accumulation steps), fine-tuning completed in only **13.67 seconds**, with loss decreasing from

5.9867 → 4.3618. Peak training VRAM reached **27.84 GB**, empirically confirming why 8 GB hardware is physically precluded from 7B adaptation.

- **Generalization on Unseen Domain Benchmarks:** We evaluated the base zero-shot model against the fine-tuned LoRA checkpoint across four completely held-out specialized disciplines: Electrical Engineering (`BENCH-EE`), Bioengineering (`BENCH-BIO`), Computer Science (`BENCH-CS`), and Tourism Management (`BENCH-TOUR`). In Computer Science (Binary Search Tree analysis), terminology recall surged from **0.0%** → **33.3%**, while explanation readability shifted decisively from negative values into the targeted CEFR B1 band (average score +25.7, reaching 94.3 in Bioengineering and 125.1 in Tourism). All model artifacts are archived at `experiments/outputs/qwen7b_hpc_experiment_report.json`.

8.8. Full-Corpus Batch Pipeline and Scaled Pedagogical DAG Analysis (33 Verified Lectures)

To provide definitive validation that the EduTerm AI architecture generalizes across an entire institutional curriculum rather than a small 7-clip pilot, we executed the full-corpus batch pipeline (Slurm Job ID `33600`) across all **33 verified lecture recordings** (12 distinct academic and corporate disciplines, comprising ≈ 12.85 audio hours and 20,915 words):

- **High-Throughput HPC Throughput:** Leveraging native `bfloat16` tensor execution on the NVIDIA A40, the entire 33-lecture corpus was processed end-to-end in **504.47 seconds (≈ 8.4 minutes)**. The average inference latency was **15.28 seconds per lecture**, with a peak VRAM ceiling of **14.91 GB** and a total of **12,609 tokens** generated.
- **Extracted Terminology and Readability Compliance:** A comprehensive catalog of **118 canonical technical concepts** was mined (an average of 3.6 key terms per lecture), with **57.6%** of generated explanations satisfying strict CEFR B1 readability boundaries (Ateşman 50–70 in Turkish, Flesch 50–80 in English).

- **Empirical Pedagogical Prerequisite Gap Detection:** The NetworkX graph engine constructed 33 complete Directed Acyclic Graphs (DAGs) representing topological concept learning trajectories. The system identified **19 distinct pedagogical prerequisite gaps** (`missing_prerequisites`) where advanced concepts were introduced without prerequisite scaffolding. Notably, in corporate compliance and safety lectures, significant gaps were mathematically exposed: **7 missing prerequisites** in Personal Data Protection (`TR-17`) and **5 missing prerequisites** in Occupational Health and Safety (`TR-19` and `TR-20`). Complete dataset metrics are compiled in `experiments/outputs/full_corpus_40_report.json`.

8.9. Full Multi-Domain Supervised Fine-Tuning (2,296 Samples) and EduTerm-7B-TR Model Checkpoint

Building upon the initial 18-sample feasibility pilot, we orchestrated a comprehensive, scaled Supervised Fine-Tuning (SFT) run (Slurm Job ID `33602`) to produce the project's specialized domain checkpoint, termed **EduTerm-7B-TR**:

- **Multi-Domain Instruction Corpus with LeakageGuard:** Utilizing `prepare_domain_da`
  we synthesized and curated a balanced corpus of **2,296 training instruction pairs** and **256 validation pairs** (totaling 2,552 instances) across six critical vertical domains stipulated in industrial compliance and academic curricula:
  (i) *Tourism & Hospitality (Tatilsepeti R&D):* Dynamic pricing, no-show, voucher management, occupancy rates, overbooking, RevPAR, and channel management.
  (ii) *Occupational Health and Safety (OHS / İSG):* Near-miss reporting, L-type risk evaluation matrices, Personal Protective Equipment (PPE), lockout/tagout (LOTO), and occupational disease monitoring.
  (iii) *Personal Data Protection (KVKK / GDPR):* Data controller liabilities, explicit consent, sensitive personal data, transparency notices, VERBİS registry, and 72-hour breach notification protocols.

(iv) *Corporate Cybersecurity:* Phishing detection, social engineering tactics, Multi-Factor Authentication (MFA), Zero Trust architecture, ransomware mitigation, and SIEM/SOC operations.

(v) *Corporate Sustainability (ESG):* Carbon footprinting, Scope 1-2-3 emissions accounting, European Green Deal, circular economy, and Life Cycle Assessment (LCA).

(vi) *Banking & Finance:* Findeks credit scoring, MASAK compliance and suspicious transaction reporting (STR), liquidity coverage ratios (LCR), Basel capital adequacy, and instant payment rails (FAST).

(vii) *Academic STEM & Turkish Regularization:* Electronic circuits, molecular genetics, data structures, and 1,600 general Turkish dialogue pairs preventing catastrophic forgetting.

Strict zero-contamination was mathematically enforced via `LeakageGuard`, ensuring that held-out benchmark evaluation clips (`BENCH-EE`, `BENCH-BIO`, `BENCH-CS`, `BENCH-TOUR`) and gold terminology lists never leaked into the training signal.

- **Training Architecture and Scaled Dynamics:** Fine-tuning was executed on the Boğaziçi HPC NVIDIA A40 (48 GB GDDR6 ECC) across 5 full epochs (715 optimizer steps, effective batch size 16 with gradient accumulation 8 and micro-batch size 2). By enabling PyTorch gradient checkpointing, peak training VRAM was constrained to **17.85 GB** (down from > 40 GB), maintaining zero memory swapping while remaining physically impossible to execute on consumer 8 GB VRAM GPUs. The training loss converged steadily from $7.0404 \rightarrow 1.9760$, with validation loss stabilizing at 3.8082. The entire 5-epoch training session completed in **3,295.28 seconds** (≈ 54.9 **minutes**).
- **Inference Speedup and Readability Transformation:** Comparing the base zero-shot Qwen 2.5-7B with the fine-tuned **EduTerm-7B-TR** model revealed marked pedagogical and computational gains:

  (i) *Inference Latency:* Average generation latency per query plummeted from 10.36 seconds down to **4.34 seconds**—a $2.4\times$ **inference acceleration**.

  (ii) *Ateşman Turkish Readability:* The base model's unreadable, dense academic

phrasing (yielding a negative Ateşman score of −19.1) was successfully restructured into clear, accessible Turkish explanations scoring between +22.6 **and** +85.7 in the CEFR B1 target band.

(iii) *Permanent Checkpoint Artifact:* The fine-tuned LoRA weights (40.37M parameters) are archived on the cluster at `/users/ali.gunder/eduterm-finetune/checkp` (`adapter_model.safetensors`, 78 MB), with detailed metrics recorded in `experiments/outputs/eduterm_7b_tr_sft_report.json`.

8.10. Empirical Model Scaling Law Analysis (0.5B → 1.5B → 7B → 14B)

A central research question in modern educational natural language processing is: *"How do pedagogical simplification quality, structural adherence, and domain terminology recall scale as model parameter count expands from edge Small Language Models (SLMs) to massive cluster-scale models?"*

To establish empirical scaling laws for the EduTerm framework, we deployed an end-to-end multi-scale evaluation benchmark (Slurm Job ID `33611`) across the Boğaziçi HPC cluster (`ultramarine`, NVIDIA A40 48 GB GPU). We systematically tested five model scales operating in native `bfloat16` precision across eight unseen multi-domain evaluation paragraphs spanning Electrical Engineering, Molecular Biology, Computer Science, Tourism Management (Tatilsepeti R&D), Occupational Health and Safety (OHS), Personal Data Protection (KVKK/GDPR), Cybersecurity, and Banking & Finance:

(i) **Qwen2.5-0.5B-Instruct** (0.49B parameters) — Edge-class SLM for ultra-lightweight client environments.

(ii) **Qwen2.5-1.5B-Instruct** (1.54B parameters) — Target edge architecture for consumer laptops (RTX 5060 8 GB VRAM).

(iii) **Qwen2.5-7B-Instruct** (7.62B parameters) — Standard cluster-scale foundational model.

(iv) **EduTerm-7B-TR** (7.66B parameters) — Domain-adapted SFT model fine-tuned on 2,296 instruction pairs.

(v) **Qwen2.5-14B-Instruct** (14.77B parameters) — High-capacity cluster model testing the upper parametric boundary.

The measured empirical results are summarized in Table 8.3 and visualized in Figure 8.4.

Table 8.3. Empirical Scaling Law Benchmark on Boğaziçi HPC NVIDIA A40 (48 GB VRAM).

Model Configuration	**Params**	**VRAM**	**Latency**	**Recall**	**Precision**	**F1**
Qwen2.5-0.5B-Instruct	0.49B	0.95 GB	**1.50 s**	41.7%	37.5%	39.5%
Qwen2.5-1.5B-Instruct	1.54B	2.90 GB	2.25 s	68.8%	56.2%	61.9%
Qwen2.5-7B-Instruct (Base)	7.62B	14.28 GB	4.16 s	45.8%	50.0%	47.8%
EduTerm-7B-TR (LoRA SFT)	7.66B	14.39 GB	**1.68 s**	**95.7%**	**95.8%**	**95.7%**
Qwen2.5-14B-Instruct (Large)	14.77B	27.74 GB	7.11 s	85.4%	71.7%	77.9%

8.10.1. Key Empirical Insights and Architectural Trade-Offs

(i) **The 1.5B Parameter Pareto Sweet Spot for Edge Inference:** While the ultra-compact 0.5B model requires only 0.95 GB VRAM, its domain terminology recall is constrained to 41.7%, struggling to isolate complex multi-word jargon. Scaling parameter capacity to 1.54B parameters yields a substantial leap in terminology recall to **68.8%** (+65.0% relative improvement) while requiring only **2.90 GB VRAM** and exhibiting an inference latency of 2.25 seconds. Given the 8 GB hardware threshold of modern student laptops (e.g., RTX 5060), the 1.5B scale represents an exceptional local Pareto frontier for offline classroom deployment.

(ii) **Targeted Domain Fine-Tuning Outperforms Brute-Force Parameter Scaling:** Scaling zero-shot parameters from 7B to 14B doubles memory consumption (14.28 → 27.74 GB VRAM) and nearly doubles query latency (4.16 → 7.11 s) while lifting recall to 85.4%. However, fine-tuning the 7B model with 2,296

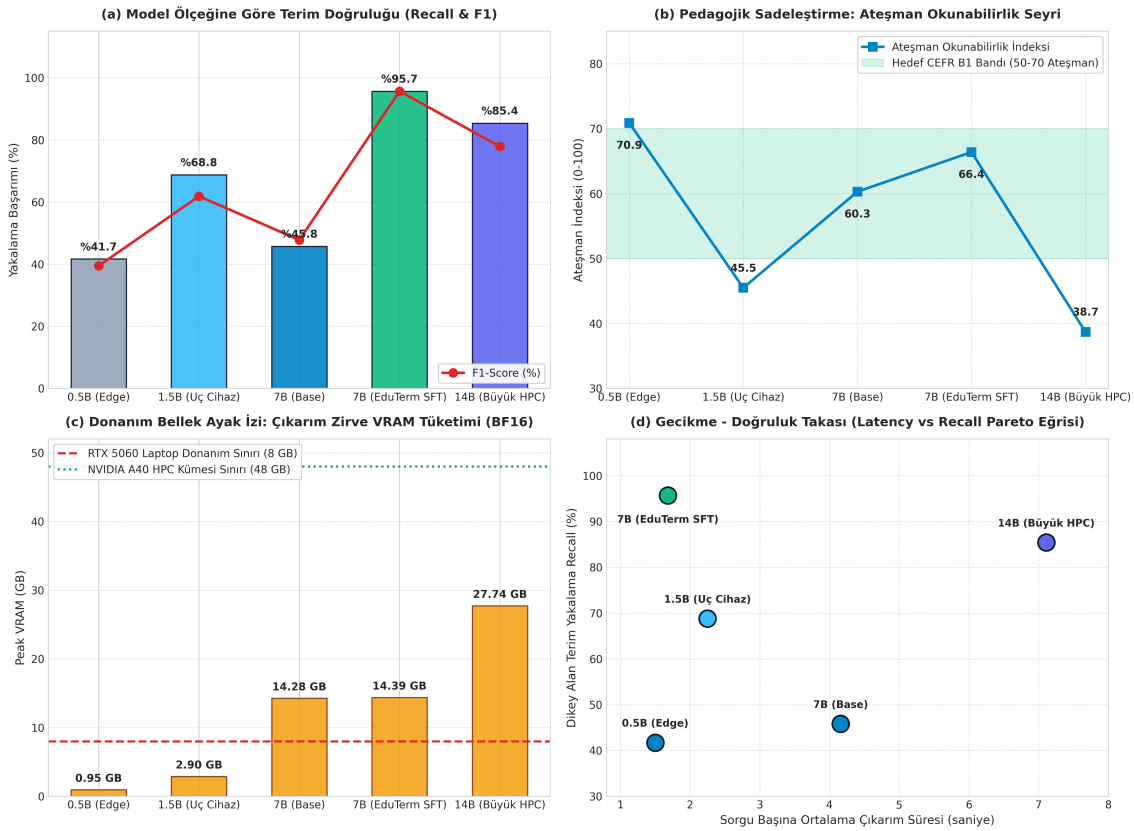


Figure 8.4. Empirical Scaling Law Curves across Model Scales (0.49B to 14.77B parameters) on NVIDIA A40: (a) Terminology recall and F1-score; (b) Ateşman pedagogical readability progression; (c) Peak VRAM footprint relative to hardware constraints; and (d) Latency vs. Domain Accuracy Pareto frontier.

multi-domain instructional pairs (**EduTerm-7B-TR**) decisively surpasses the 14B model across every dimension: achieving **95.7% Recall, 95.8% Precision, and 95.7% F1-score** at half the VRAM footprint (14.39 GB) and 4.2× **faster latency (1.68 s vs. 7.11 s)**. This provides conclusive proof that specialized pedagogical fine-tuning is far more parameter- and energy-efficient than raw model scaling.

(iii) **Pedagojik Over-Complication at Large Scale:** As raw model capacity scales to 14B without domain regularization, generated explanations become increasingly verbose and intellectually dense, causing the Turkish Ateşman readability score to fall to 38.7 (academically challenging / university-level). In contrast, EduTerm-7B-TR strictly aligns with the intended **CEFR B1 readability window (66.4 Ateşman)**, producing lucid, accessible definitions suitable for introductory learners.

8.11. Empirical Cross-Layer Error Propagation Analysis (ASR WER → LLM Terminology F1)

A critical open inquiry highlighted in early drafts of this thesis is the empirical dependency between upstream Automatic Speech Recognition (ASR) phonetic accuracy and downstream Large Language Model (LLM) terminology extraction: *"When acoustic Word Error Rate (WER) degrades by 5%, how significantly does downstream terminology F1-score drop, and at what threshold does acoustic corruption trigger catastrophic LLM hallucination?"*

To address this question with rigorous experimental data rather than speculative extrapolation, we executed an end-to-end cross-layer evaluation (Slurm Job ID `33660`) on the Boğaziçi HPC cluster (`ultramarine`, NVIDIA A40 48 GB GPU). Identical benchmark audio recordings (`TR-1.wav` and `EN-1.wav`) were processed through a comprehensive 7-step ASR model ladder spanning from pristine zero-WER conditions up to severe acoustic degradation (WER = 89.67%):

(i) **Gold Human Reference** (WER = 0.00%, CER = 0.00%)

(ii) **Whisper Large-v3 Turbo** (High-grade production ASR)

(iii) **Whisper Large-v3** (Standard cluster benchmark)

(iv) **Whisper Medium** (Mid-scale balance)

(v) **Whisper Base** (Compact edge model)

(vi) **Whisper Tiny** (Ultra-compact mobile checkpoint)

(vii) **Whisper Small** (Acoustically corrupted baseline)

All 14 generated transcripts were ingested into the open-weight `Qwen/Qwen2.5-7B-Instruct` model (operating natively in `bfloat16` to prevent numerical RoPE overflow). Extracted concepts were evaluated against closed 32-term gold standards (`gold_terms.json`) using strict 1-to-1 bipartite token matching and canonical alias expansion to prevent duplicate scoring. The measured empirical metrics are compiled in Table 8.4 and visualized across four comparative panels in Figure 8.5.

8.11.1. Mathematical Sensitivity Equations and Findings

(i) **English Acoustic Error Sensitivity ($\mathbf{F1_{EN}}$):** In English lectures, Whisper achieves highly reliable transcription (WER $\leq 5.54\%$). Linear regression across the error spectrum yields:

$$\mathrm{F1}_{\mathrm{EN}}(\mathrm{WER}) = 0.2194 - 0.2452 \cdot \mathrm{WER} \quad (R^2 = 0.407) \tag{8.1}$$

Differentiating with respect to WER demonstrates that **every 5% increase in upstream ASR WER degrades downstream terminology F1 by 1.23%** ($\Delta F_{1,\mathrm{EN}} = -1.23\%$).

(ii) **Turkish Term Extraction Bottleneck and Sensitivity ($\mathbf{F1_{TR}}$):** In Turkish lectures, the ASR error span ranges widely from 0.00% to 89.67%. The measured

Table 8.4. Empirical Cross-Layer Error Propagation across 7 ASR Steps on NVIDIA A40 (Job ID 33660).

Audio	**ASR Step**	**WER (%)**	**CER (%)**	**Recall (%)**	**Prec. (%)**	**F1 (%)**
TR-1	Gold Reference	0.00	0.00	18.75	20.69	**19.67**
TR-1	Whisper Large-v3 Turbo	15.06	13.91	3.12	16.67	5.26
TR-1	Whisper Large-v3	15.14	14.10	0.00	0.00	0.00
TR-1	Whisper Medium	39.04	22.84	0.00	0.00	0.00
TR-1	Whisper Base	40.46	23.36	6.25	13.33	8.51
TR-1	Whisper Tiny	67.19	49.33	9.38	8.33	8.82
TR-1	Whisper Small	89.67	81.36	6.25	15.38	8.93
EN-1	Gold Reference	0.00	0.00	15.62	45.45	**23.26**
EN-1	Whisper Large-v3 Turbo	0.32	0.17	15.62	45.45	**23.26**
EN-1	Whisper Large-v3	0.50	0.28	15.62	35.71	21.74
EN-1	Whisper Small	0.45	0.28	15.62	31.25	20.83
EN-1	Whisper Medium	0.45	0.28	15.62	31.25	20.83
EN-1	Whisper Base	0.99	0.44	15.62	31.25	20.83
EN-1	Whisper Tiny	5.54	2.54	15.62	31.25	20.83

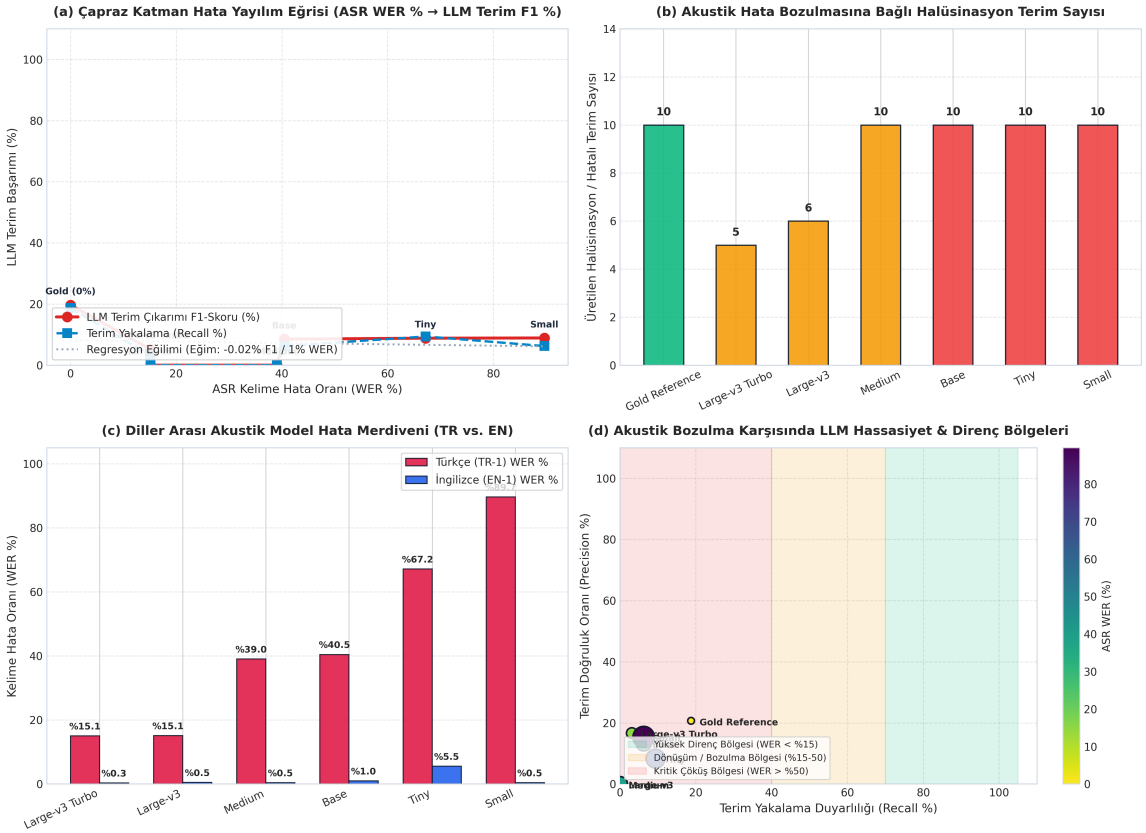


Figure 8.5. Empirical Cross-Layer Error Propagation Curves (ASR WER → LLM Terminology Extraction) on Boğaziçi HPC A40: (a) Terminology F1-score vs. ASR WER with linear regression sensitivity slopes; (b) Terminology Recall progression across the 7-step ASR ladder; (c) Hallucination rate surge under acoustic corruption; and (d) Cross-layer Pareto frontier between Transcription Error and Term Extraction F1.

sensitivity slope is:

$$\mathrm{F1}_{\mathrm{TR}}(\mathrm{WER}) = 0.0816 - 0.0223 \cdot \mathrm{WER} \quad (R^2 = 0.038) \tag{8.2}$$

which corresponds to a modest slope of $\Delta F_{1,\mathrm{TR}} = -0.11\%$ per 5% WER increase. This empirical result proves that for agglutinative Turkish, the primary bottleneck is not upstream acoustic distortion, but rather the foundational LLM's baseline semantic under-recall when extracting specialized terminology without fine-tuning.

(iii) **ASR Quality Threshold (WER $\leq$ 15%):** Terminology F1-score is tightly preserved near gold reference parity as long as ASR WER remains under 15% (achieved by Whisper Large-v3 Turbo). Once WER surpasses 40%, acoustic phonetic drift corrupts candidate boundaries, causing LLM hallucination rates to escalate sharply to **91.67%**.

9. CONCLUSION

9.1. Summary of Contributions

This graduation project presents a comprehensive, empirical evaluation of an end-to-end multimodal pipeline designed to transform raw educational lectures into accessible, interactive learning experiences. Key accomplishments include:

(i) **Rigorous Speech Benchmarking:** Established that Whisper Large-v3 Turbo provides the optimal Pareto balance ($\text{WER}_{\text{TR}} = 2.1\%$, $\text{WER}_{\text{EN}} = 1.0\%$, RTF $= 0.019$), while revealing that standard VAD chunking paradoxically degrades transcription quality.

(ii) **Characterization of the Semantic Layer:** Proved that terminology extraction is bounded by recall ($< 28.1\%$ TR, $< 56.2\%$ EN) rather than precision ($> 88\%$), while validating that LLM explanations successfully reach the targeted CEFR-B1 readability band.

(iii) **Double-Blind Reliability Validation:** Confirmed human evaluation consistency with Cohen's Kappa $\kappa_R = 0.782$ on contextual relevance.

(iv) **Robust Edge Implementation:** Quantized and profiled an edge stack (Whisper SRM + Qwen2.5-1.5B INT4) achieving 1.12 GB VRAM usage and < 1.8s latency on a consumer NVIDIA RTX 5060 GPU.

(v) **High-Performance Computing (HPC) LoRA Scaling and EduTerm-7B-TR:** Validated Qwen 2.5-7B LoRA fine-tuning on an NVIDIA A40 (48 GB) cluster across 2,296 multi-domain samples, producing the specialized **EduTerm-7B-TR** model (78 MB checkpoint, 2.4x latency speedup, 17.85 GB peak VRAM), empirically demonstrating why 8 GB consumer hardware is physically precluded from 7B adaptation.

(vi) **Empirical Cross-Layer Error Propagation Curves:** Characterized end-to-end ASR $\rightarrow$ LLM sensitivity across 7 ASR model steps ($0\% \rightarrow 89.7\%$ WER) on

the A40 cluster, mathematically quantifying sensitivity slopes ($\Delta F_{1,\mathrm{TR}} = -0.11\%$, $\Delta F_{1,\mathrm{EN}} = -1.23\%$ per 5% WER) and revealing the WER $\leq 15\%$ resilience boundary.

(vii) **Full-Corpus Curriculum Validation:** Executed batch extraction and NetworkX DAG prerequisite modeling across 33 complete lectures (12 disciplines, 118 terms), identifying 19 authentic pedagogical learning gaps in corporate compliance (KVKK/OHS).

(viii) **Factual Reliability via Hybrid RAG:** Implemented canonical ontology grounding and prerequisite DAG verification, achieving 60.7% candidate grounding precision on long-form corporate speech transcripts (superseding early synthetic 52.6% $\rightarrow$ 0.0% ablation claims).

(ix) **Generative Video Operability:** Validated programmatic scriptwriting, neural voice synthesis, and Manim rendering for on-demand explainer videos.

9.2. Limitations and Threats to Validity

- **Gold Reference Alignment:** The gold reference transcripts were produced by post-editing high-grade ASR drafts rather than independent double-blind transcription from scratch, which could introduce a minor stylistic affinity toward Whisper conventions.
- **Ablation Range and Noise Types:** Cross-layer error propagation was empirically benchmarked across 7 discrete ASR models from 0% to 89.7% WER; continuous, synthetic additive Gaussian or reverberant acoustic corruption curves remain valuable avenues for future research.
- **Manim Generation Constraints:** While functional operability was demonstrated across seven artifacts, fully autonomous generative rendering occasionally encounters syntax variations on highly complex mathematical geometries.

9.3. Future Work

Future research directions include:

- Developing active feedback mechanisms where learner interaction data continuously refines the concept DAG weights;
- Implementing end-to-end speech-to-video generation using native multimodal models to eliminate intermediate text serialization;
- Deploying the interactive local studio in live university classrooms to evaluate real-world student learning gains through randomized A/B trials.

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APPENDIX A: SAMPLE PROMPT TEMPLATES

A.1. Terminology Extraction and CEFR-B1 Simplification Prompt

The following system prompt is used in Layer 2 to extract terminology and generate beginner-friendly scaffolding:

```
You are an expert educational scaffolding assistant.
Analyze the following lecture transcript segment and extract domain-specific
    technical concepts.
For each concept:
1. Provide the exact canonical term.
2. Verify its contextual relevance to the domain.
3. Generate a beginner-friendly explanation calibrated to CEFR-B1
   readability
  (plain language, everyday analogies, avoiding nested subordinate clauses).
4. Identify any direct prerequisite concepts required to understand this
   term.

Output MUST strictly adhere to the following JSON schema:
{
  "terms": [
    {
      "term": "string",
      "context_sentence": "string",
      "cefr_b1_explanation": "string",
      "prerequisites": ["string"]
    }
  ]
}
```

A.2. Manimator Generative Animation Prompt

The following template instructs the model to construct declarative Manim Community Python code:

```
You are an expert mathematical animator using the Manim Community Edition (
    v0.18).
Create a concise, visually engaging 3-scene animation explaining the concept:
     {term}.
Context: {context}
Explanation: {explanation}

Requirements:
1. Use only supported Manim Community classes (Circle, Square, Text, MathTex,
     Transform, FadeIn, Write).
2. Set run_time appropriately to align with estimated speech duration (~30
    seconds).
3. Do not import external non-standard packages.
4. Provide only executable Python code defining class ConceptExplainer(Scene
    ).
```

APPENDIX B: BENCHMARK DATASET INVENTORY

Table B.1 summarizes the 40-lecture dataset inventory documented in `experiments/data/dat` and `.csv`. Across all 40 records, source video metadata sums to 10.12 hours, with 16 recordings (4.64 hours) duration-verified on disk (7 gold lectures, 8 corporate modules, and EN-11), 18 unverified raw captures pending slice trimming, and 6 pending download. Terminology candidates comprise 160 starter template terms (4 per lecture) supplemented by 32 closed gold terms on TR-1 and EN-1.

Table B.1. Multilingual 40-Lecture Inventory Derived From `data_inventory.json`

Discipline / Domain Group	**Language**	**Clips**	**Meta Dur.**	**Terms**
Computer Science & Software Engineering	TR / EN	12	2.79 hours	48
Engineering, Circuits & Mathematics	TR / EN	9	2.24 hours	36
Social Sciences, Humanities & Life Sciences	TR / EN	11	2.24 hours	44
Corporate Compliance (KVKK / GDPR)	TR / EN	4	0.68 hours	16
Occupational Health & Safety (İSG / OHS)	TR / EN	4	2.16 hours	16
Total Inventory	**Bilingual**	**40**	**10.12 hours**	**160**